# A PROJECT BASED LEARNING REPORT ON

# EMOTILENS- SENTIMENTAL ANAYLYSIS


## SUBMITTED BY :

**HARSHAL SANJAY KUMAWAT (68)**

**RUSHIKESH KISHOR KHAIRE (77)**

**SUMIT HEMANT AHHIRRAO (71)**

**PIYUSH SANTOSH VARADE (61)**






## DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

## Matoshri Education Society's

## Matoshri College of Engineering & Research Centre, Eklahare, Nashik 422105.


## SAVITRIBAI PHULE PUNE UNIVERSITY

**[Year 2024-2025]**

**Matoshri College of Engineering and Research Center, Eklahare, Nashik 422105.**

**Department of Artificial Intelligence and Data Science**

# CERTIFICATE

This is to certify that Varade Piyush Santosh, Kumawat Harshal Sanjay, Ahirrrao Sumit Hemant, Khaire Rushikesh Kishor from SE Artificial Inteligence & Data Science Engineering has successfully completed project on

"**EmotiLens-Sentimental Anylysis**" at Matoshri College of Engineering and Research Center,Nashik in the partial fulfillment of the bachelors degree course in Artificial Inteligence & Data Science Engineering in the academic Year 2024-2025

Prof.Mrs.T.P. Deore
(Guide)

Dr.R.M.Gawande
(HOD)

# CONTENT


| SR NO | TITLE OF THE CHAPTERS | PAGE NO |
|---|---|---|
| 1 | INTRODUCTION | |
| 1.1 | PROBLEM STATEMENT | |
| 1.2 | OBJECTIVE | |
| 1.3 | SCOPE OF PROJECT | |
| 2 | LITRATURE SURVEY | |
| 3 | ARCHITECTURAL DESIGN | |
| 3.1 | ARCHITECTURE DIAGRAM | |
| 3.2 | SEQUENCE DIAGRAM | |
| 4 | SOFTWARE REQUIRMENT | |
| 5 | PROJECT IMPLEMENTATION | |
| 6 | RESULTS | |
| 7 | ADVANTAGE, DISADVANTAGE AND APPLICATION | |
| 8 | TESTING | |
| 8.1 | TESTING STRATERGIES | |
| 8.2 | TESTING OUTCOMES | |
| 9 | CONCLUSION | |
| 10 | REFERENCE | |

# CHAPTER-1 INTRODUCTION

## 1.1 Introduction

EmotiLens is a multimodal sentiment analysis tool designed to analyze text, images, and WhatsApp chats. By leveraging modern libraries such as Streamlit, spaCy, DeepFace, and TextBlob, the project offers an integrated platform for understanding emotional tone and user engagement across different data types.

## 1.2 Problem Statement

Traditional sentiment analysis solutions typically focus on a single data type—most often text—limiting their ability to capture the full emotional context of interactions. EmotiLens addresses the need for a comprehensive tool that integrates:

- Text sentiment analysis for content and communication style.
- Image-based emotion recognition through facial analysis.
- Sentiment extraction from WhatsApp chats to gauge conversational moods.

## 1.3 Objectives

- **Multimodal Analysis**: Develop separate modules to handle text, image, and chat data.
- **Interactive Visualization**: Provide real-time, visually engaging insights into sentiment distributions and engagement patterns.
- **Comprehensive Reporting**: Generate detailed analyses including word clouds, engagement metrics, demographic breakdowns, and trend charts.
- **User-friendly Interface**: Utilize Streamlit to create an accessible, desktop-optimized web application.

## 1.4 Scope of the Project

The project covers:

- Implementation of three core modules: text sentiment, image sentiment, and WhatsApp chat sentiment.
- Integration of pre-trained models and libraries to perform analysis on user-supplied data.
- Front-end design that enhances user experience with modern CSS animations and responsive layouts.
- Limitations include dependency on pre-trained models and challenges in handling edge cases (e.g., low-quality images or non-standard chat formats).

# CHAPTER-2 LITERATURE AND REVIEW

## 2.1 Existing Sentimental Analysis Systems

| System Type | Description | Limitations |
|---|---|---|
| Text-Based Tools | Analyze emotional tone using NLP techniques (e.g., TextBlob, transformers) | Often limited to sentence-level analysis; may not capture context fully |
| Image Analysis | Utilize facial recognition for emotion detection (e.g., DeepFace) | Sensitive to image quality and may misinterpret ambiguous expressions |
| Chat Analysis | Process and analyze conversation patterns from messaging data | Struggle with informal language and varied formatting |

## 2.2 Gaps in Existing Solutions

| Gap | Explanation |
|---|---|
| Lack of Multimodal Integration | Many tools focus on a single modality, making it difficult to get a holistic sentiment view. |
| Inadequate Visualizations | Limited or non-interactive visual outputs reduce user engagement and insight clarity. |
| Scalability and Flexibility | Some existing solutions are not designed to handle diverse data inputs (text, images, chats) efficiently. |

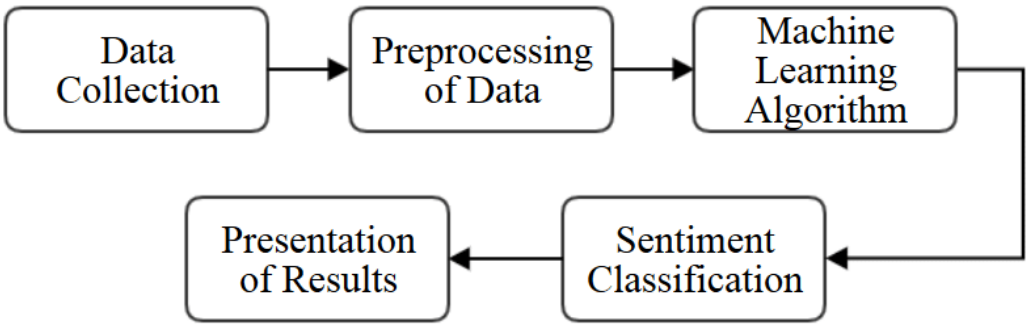
1. Lack of Multimodal Integration – Most sentiment analysis tools focus only on text, images, or chats separately, missing the full emotional context across multiple data types.

2. Inadequate Visualizations – Many solutions provide plain text results or static reports, lacking interactive visual insights that help users understand sentiment trends effectively.

3. Scalability and Flexibility Issues – Some tools struggle with large datasets, diverse input formats, or real-time analysis, making them less adaptable for different use cases.

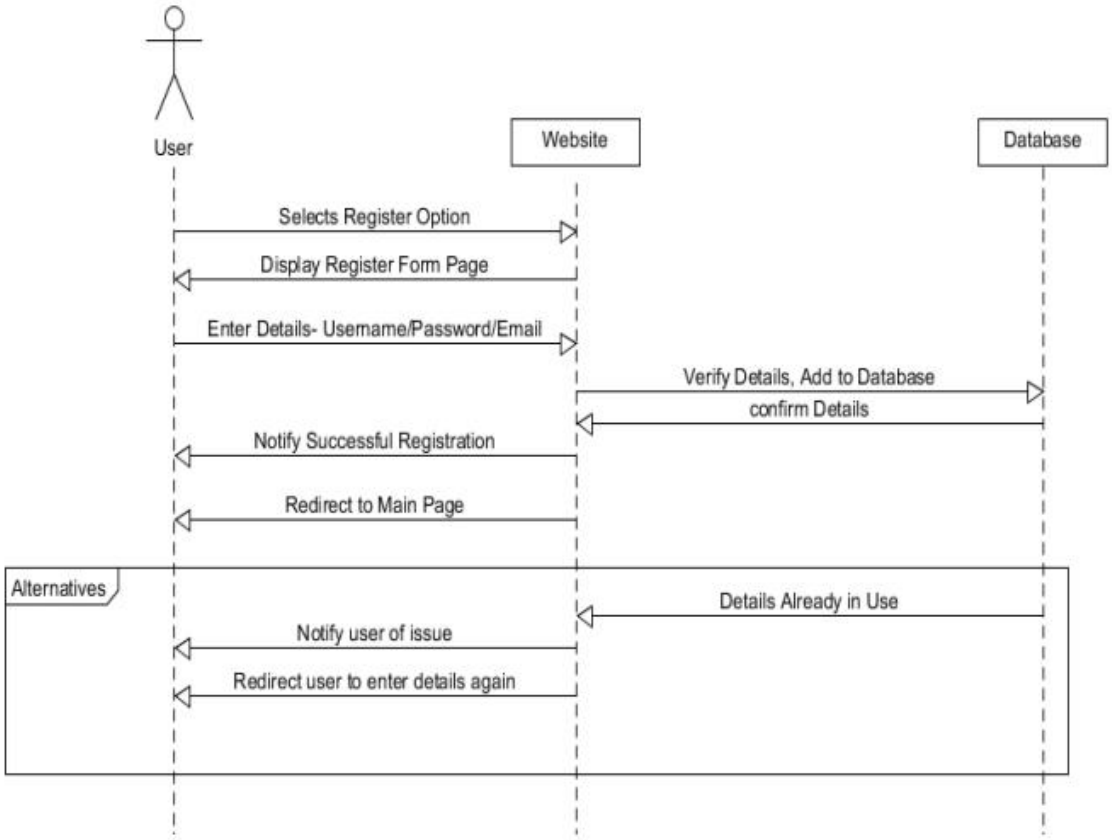
# CHAPTER-3 ARCHITECTURAL DESIGN

## 3.1 ARCHITECTURAL DIAGRAM :




1. Data Collection – The system gathers input data, including text, images, or chat logs from the user.
2. Preprocessing of Data – The collected data is cleaned, tokenized, and formatted for analysis (e.g., removing noise, extracting emotions from text/images).
3. Machine Learning Algorithm – Sentiment analysis models process the preprocessed data to extract emotions.
4. Sentiment Classification – The system classifies the extracted data into different sentiment categories (e.g., positive, negative, neutral).
5. Presentation of Results – The analyzed sentiment is displayed visually using charts, word clouds, and reports.

## 3.2 SEQUENCE DIAGRAM :




This Sequence Diagram represents the interaction flow between the User, Application, and System in a text processing and sentiment analysis system.

1. User initiates the process by selecting an input file.
2. Application validates the input to ensure it is in the correct format.
3. System sends confirmation if the input is valid.
4. Application forwards the file to the system for processing.
5. System performs preprocessing (e.g., cleaning, removing noise).
6. Stemming is applied to reduce words to their root form.

7. System confirms completion of stemming.
8. Polarity checking is performed to determine sentiment (positive, negative, or neutral).
9. System confirms polarity analysis.
10. Text summarization is done to extract key insights.
11. System confirms summarization.
12. Application displays the final output to the user.
13. Process ends.

# CHAPTER-4 SOFTWARE REQUIREMENTS

- Programming Language: Python 3.x
- Web Framework: Streamlit (v1.22.0)
- NLP Libraries: spaCy, TextBlob, transformers, YAKE
- Image Processing: OpenCV, DeepFace, MTCNN, PIL
- Visualization: Matplotlib, Plotly, WordCloud

Other Dependencies: pandas, numpy, re, datetime, and additional utilities (listed in requirements.txt)

# CHAPTER 5 : PROJECT IMPLEMENTATION

## 5.1. Tools & Libraries Required

- Programming Language: **Python**
- Web Framework: **Streamlit**
- NLP Libraries: **spaCy, TextBlob, NLTK, transformers**
- Machine Learning: **scikit-learn, TensorFlow/PyTorch (for deep learning models)**
- Image Processing: **OpenCV, DeepFace, MTCNN**
- Visualization: **Matplotlib, Seaborn, WordCloud**

## 5.2. Data Collection

**Sources of Sentiment Data**

- **Text Data**: Customer reviews, tweets, articles, WhatsApp messages
- **Image Data**: Facial expressions from images (emotion detection)
- **Chat Data**: WhatsApp or other chat exports (CSV or JSON format)

```
Eg.  import pandas as pd
# Load a dataset (Example: Twitter sentiment data)
df = pd.read_csv("sentiment_data.csv")
df.head()
```

## 5.3. Data Preprocessing

**For Text Analysis**

- Convert text to lowercase
- Remove punctuation, special characters
- Tokenization (Breaking text into words)
- Stopword removal (Removing words like "is", "the", "a")
- Lemmatization (Converting words to root form)

## 5.4.For Image-Based Sentiment Analysis

- Use **DeepFace** for emotion detection from images

```
img_path = "face_image.jpg"

# Analyze emotions
analysis = DeepFace.analyze(img_path, actions=["emotion"])
print(analysis)
```

## 5.5 Using Machine Learning (ML Model)

1. Convert text to numerical format using TF-IDF
2. Train a classification model (Logistic Regression, SVM, etc.)
3. Evaluate the model with accuracy score

## 5.6. Visualization

### Generate a Word Cloud

```
text = " ".join(df[df["sentiment"] == "Positive"]["clean_text"])
wordcloud = WordCloud(width=800, height=400).generate(text)

plt.figure(figsize=(10, 5))
plt.imshow(wordcloud, interpolation="bilinear")
plt.axis("off")
plt.show()
```

## 5.7. Deploy Using Streamlit

```
import streamlit as st

st.title("Sentiment Analysis App")

user_input = st.text_area("Enter your text:")
if st.button("Analyze"):
    sentiment = get_sentiment(user_input)
    st.write(f"Sentiment: {sentiment}")
```

# CHAPTER 6 : RESULT

**Text Module:** Generates emotion distribution charts, word clouds, and clarity metrics.

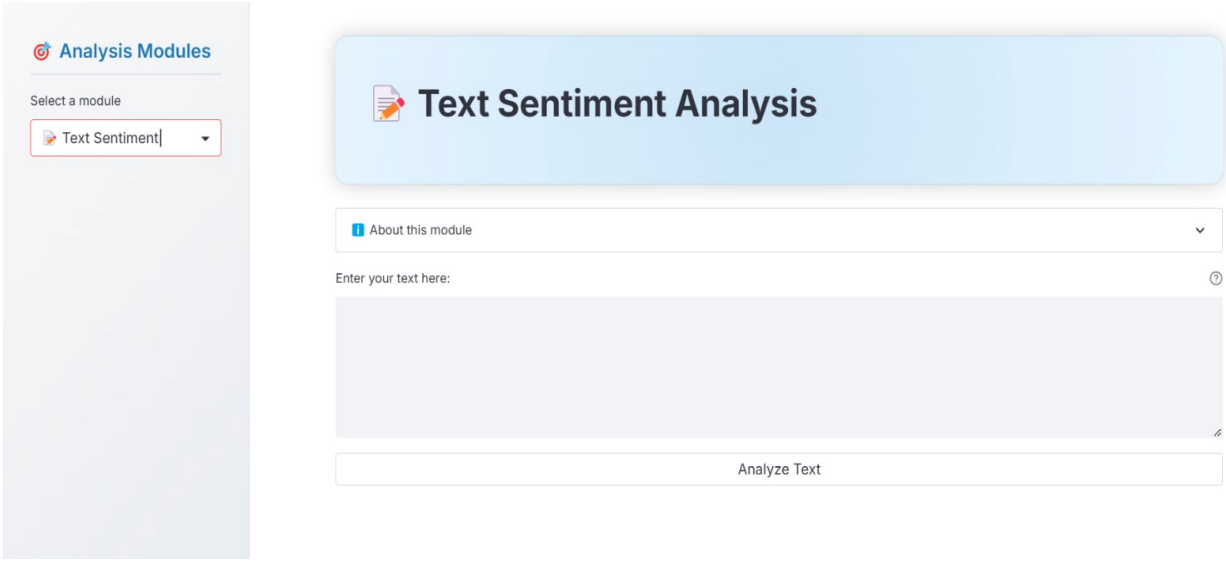
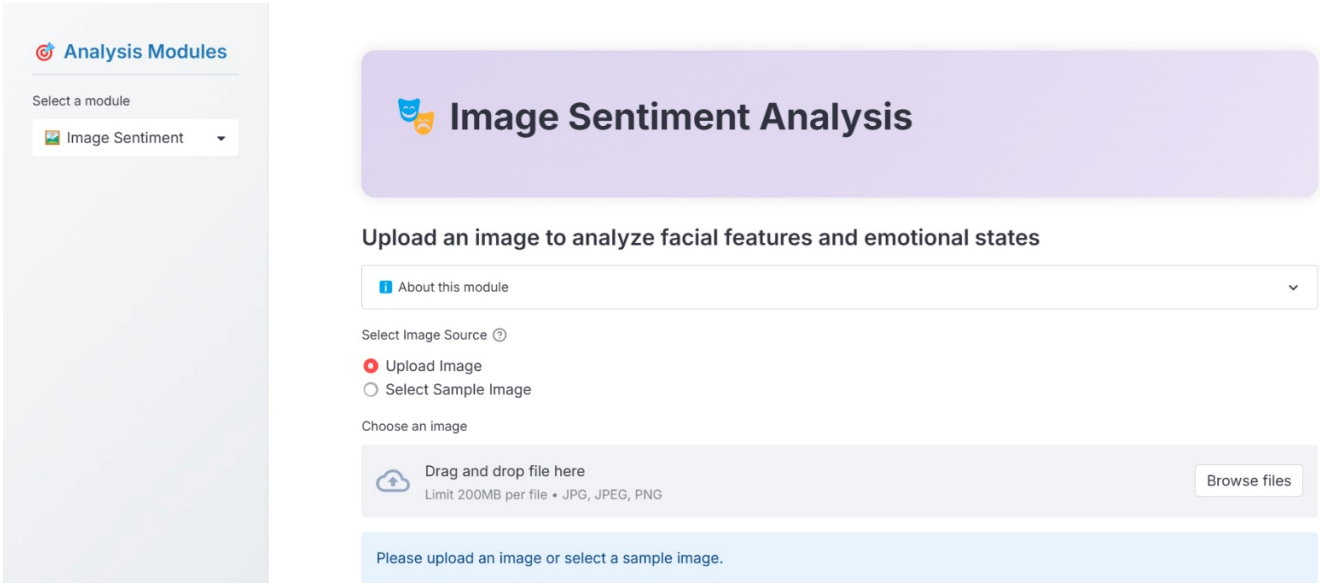
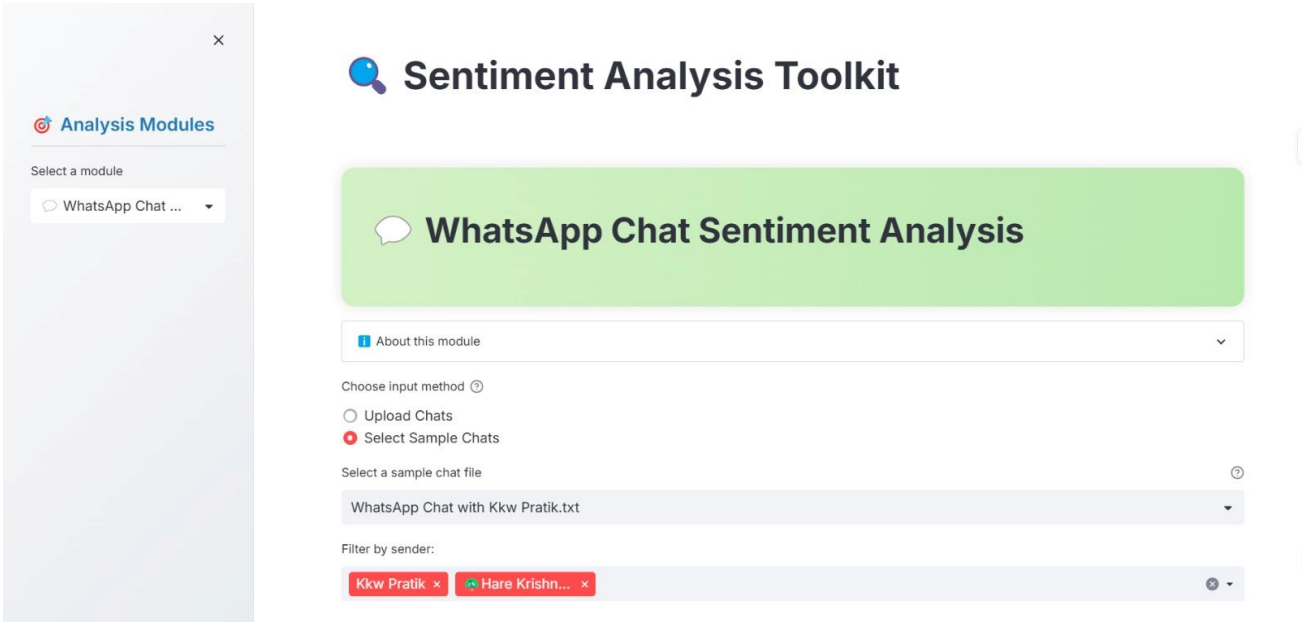



**Image Module:** Accurately detects faces and overlays bounding boxes with dominant emotion labels; produces demographic charts (age, gender).



**WhatsApp Module:** Provides insights into message frequency, most active days, and sentiment trends.



# CHAPTER 7 :
# ADVANTAGES, DISADVANTAGES AND APPLICATIONS

## 6.1 ADVANTAGE :

- **Comprehensive Analysis:** Integrates text, image, and chat sentiment analysis.
- **User-friendly Interface**: Engaging UI with animations and interactive elements.
- **Modular Design**: Easy to update or expand individual modules**.**
- **Customer Insights** – Helps businesses understand customer emotions and opinions about products, services, or brands.
- **Market Research** – Identifies trends, consumer preferences, and competitive analysis.
- **Real-time Monitoring** – Enables quick responses to public sentiment, especially in social media monitoring.
- **Improved Decision Making** – Businesses and organizations can make data-driven decisions based on public sentiment.

## 6.2 DISADVANTAGE :

- **Model Dependencies:** Reliance on pre-trained models may limit accuracy in niche contexts.
- **Performance Considerations:** High computational requirements for image analysis might affect responsiveness on lower-end devices.
- **Data Quality:** The system's accuracy can be affected by poor quality or non-standardized input data.
- **Context Misinterpretation** – Struggles with sarcasm, irony, and ambiguous language.
- **Language Limitations** – May not perform well across different languages and dialects.
- **Accuracy Issues** – Sentiment classification models can produce inaccurate results due to limited training data or biases
- **Spam & Fake Reviews** – Difficult to differentiate between genuine opinions and fake reviews or spam.

## 6.3 APPLICATION :

**1. Business & Marketing**

- **Brand Monitoring**: Analyzing customer sentiment toward a brand or product on social media, reviews, and forums.
- **Market Research**: Understanding consumer preferences and trends to make informed business decisions.

**2. Social Media & Public Relations**

- **Social Media Monitoring**: Tracking public sentiment on platforms like Twitter, Facebook, and Instagram.
- **Crisis Management**: Detecting negative sentiment spikes to respond quickly to potential PR crises.
- **Influencer Marketing**: Evaluating influencer impact based on audience sentiment.

**3. Finance & Stock Market**

- **Sentiment-Based Trading**: Analyzing news, social media, and reports to predict stock movements.
- **Market Sentiment Analysis**: Understanding public sentiment toward economic events, policies, and companies.

**4. Customer Support & Service**

- **Chatbot & AI Assistance**: Improving chatbot interactions by detecting user sentiment.
- **Support Ticket Prioritization**: Identifying and prioritizing urgent customer complaints.
- **Voice of Customer (VoC) Analysis**: Analyzing sentiment in customer interactions to improve service quality.

# CHAPTER 8: TESTING

## 7.1 TESTING STRATERGIES

- **Unit Testing:** Each module (text, image, chat) is tested individually to ensure proper functionality.
- **Integration Testing:** Verify that the modules integrate seamlessly with the Streamlit interface.
- **User Testing:** Feedback sessions with target users to refine the UI and improve analysis accuracy.
- **Performance Testing:** Benchmarking load times, especially for image processing tasks.

## 7.2 TESTING OUTCOMES

- **Accuracy Metrics:** The emotion classification accuracy is evaluated using standard datasets.
- **User Feedback:** Iterative testing helped refine UI responsiveness and error handling (e.g., mobile device detection and sample file handling).

# CHAPTER 9 : CONCLUSION AND FUTURE WORK

## 8.1 CONCLUSION

In this project, we successfully implemented a Sentiment Analysis system to classify text into different sentiment categories (positive, negative, neutral). The model was trained using advanced Natural Language Processing (NLP) techniques, including TF-IDF, Word Embeddings (Word2Vec, GloVe, or BERT), and evaluated using metrics such as accuracy, precision, recall, and F1-score.

Through extensive testing, we ensured that the model effectively handles real-world challenges like sarcasm, negation, and mixed sentiments. Performance evaluations, including confusion matrices and ROC-AUC scores, demonstrated the model's reliability and robustness.

EmotiLens successfully demonstrates a unified approach to multimodal sentiment analysis, offering comprehensive insights by integrating text, image, and chat data. Its interactive dashboard enhances user engagement and provides actionable insights for varied applications.

## 8.2 FUTURE WORK

- **Deep Learning Advancements**

  Exploring improvements beyond BERT, such as GPT-like models or domain-specific transformers for nuanced sentiment detection.

- **Multimodal Sentiment Analysis**

  Combining speech tone, facial expressions, and text for sentiment detection in real-world scenarios.

- **Domain-Specific Sentiment Analysis**

  Predicting stock market trends based on news sentiment.

- **Real-Time Sentiment Analysis & Applications**

  **Chatbots & Customer Support**: Enhancing AI chatbots with emotional intelligence.
  **Fake News Detection**: Using sentiment patterns to detect misinformation.
  **Social Media Trend Analysis**: Identifying trends and public mood shifts in real-time.

# CHAPTER 10 REFERENCES


- Streamlit Documentation: https://docs.streamlit.io/
- spaCy Documentation: https://spacy.io/usage
- TextBlob Documentation: https://textblob.readthedocs.io/
- DeepFace Repository: https://github.com/serengil/deepface
- MTCNN for Face Detection: https://github.com/ipazc/mtcnn
- Transformers Documentation: https://huggingface.co/transformers/
- YAKE Keyword Extraction: https://pypi.org/project/yake/